



Nagar Yuwak Shikshan Sanstha's

## Yeshwantrao Chavan College of Engineering

(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)

Hingna Road, Wanadongri, Nagpur - 441 110

NAAC A++

Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: [www.ycce.edu](http://www.ycce.edu)

### Department of Computer Technology

Vision Statement of the Department.

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission Statement of the Department.

To nurture computer engineering professionals by offering Dynamic learning environment that encourages problem-solving using emerging technology, National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration & Holistic, ethical and sustainable development for lifelong success.



### Session 2026-2027

|                                         |                                         |
|-----------------------------------------|-----------------------------------------|
| <b>Vision:</b> Dream of where you want. | <b>Mission:</b> Means to achieve Vision |
|-----------------------------------------|-----------------------------------------|

**Program Educational Objectives of the program (PEO):** (broad statements that describe the professional and career accomplishments)

|      |                             |                                                 |                                                                        |
|------|-----------------------------|-------------------------------------------------|------------------------------------------------------------------------|
| PEO1 | <b>Preparation</b>          | <b>P: Preparation</b>                           | <b>Pep-CL abbreviation</b><br>pronounce as Pep-si-IL<br>easy to recall |
| PEO2 | <b>Core Competence</b>      | <b>E: Environment</b><br>(Learning Environment) |                                                                        |
| PEO3 | <b>Breadth</b>              | <b>P: Professionalism</b>                       |                                                                        |
| PEO4 | <b>Professionalism</b>      | <b>C: Core Competence</b>                       |                                                                        |
| PEO5 | <b>Learning Environment</b> | <b>L: Breadth (Learning in diverse areas)</b>   |                                                                        |

**Program Outcomes (PO):** (statements that describe what a student should be able to do and know by the end of a program)

#### Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

**PSO Keywords:** Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

**Integrity:** I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

**Name and Signature of Student and Date**

(Signature and Date in Handwritten)



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|          |         |                 |                    |
|----------|---------|-----------------|--------------------|
| Session  | 2026-27 | Course Name     | Web Technology Lab |
| Semester | 3       | Course Code     | 23CT1301           |
| Roll No  | 13      | Name of Student | Juhi Parwani       |

|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Practical Number      | 1(A)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Course Outcome        | <ol style="list-style-type: none"><li>1. Understand various internet technologies.</li><li>2. Design the web pages using HTML and CSS.</li><li>3. Implement the XML technology to store the data.</li><li>4. Develop the interactive web pages using JavaScript.</li></ol>                                                                                                                                                                                                                                                                                                                                                                               |
| Aim                   | To develop an interactive calculator using HTML, CSS and JavaScript events and DOM manipulation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Problem Definition    | Build an interactive calculator that responds to <b>button clicks and keyboard input</b> and dynamically updates the result using JavaScript event handling. The calculator should accept numbers from buttons, perform addition, subtraction, multiplication and division, update the display dynamically, and provide a <b>Clear (C)</b> function. It should also support keyboard input for numbers and important operations.                                                                                                                                                                                                                         |
| Theory<br>(100 words) | An event is an action that happens on a webpage, such as a mouse click or keyboard key press. JavaScript event handling allows a webpage to respond to these actions. The <code>addEventListener()</code> method is used to listen for events such as <code>click</code> and <code>keydown</code> . In this practical, events are used to operate calculator buttons and keyboard input. JavaScript functions are used to perform calculations and update the DOM dynamically. The <code>Number()</code> function converts text values into numbers, while <code>if</code> and <code>else if</code> statements select the required arithmetic operation. |



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Procedure and Execution

(100 Words)

Step for Implementation:

1. Create a project folder named **Calculator**.
2. Create **index.html**, **style.css**, and **script.js**.
3. Create the calculator display using an input element.
4. Create buttons for numbers and operators.
5. Give unique IDs to the buttons.
6. Use **getElementById()** to access HTML elements.
7. Use **addEventListener()** for button click events.
8. Create a function to add numbers to the display.
9. Store the first number and selected operator.
10. Create a function to perform calculations.
11. Use **if** and **else if** for **+**, **-**, **\***, and **/**.
12. Display the calculated result.
13. Create a Clear function.
14. Add keyboard event handling using **keydown**.
15. Test the calculator using both buttons and keyboard input.

Code:

```
<!DOCTYPE html>
<html>

<head>
  <title>My Calculator</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <h1>My Calculator</h1>

  <input type="text" id="display" readonly>

  <div>
    <button id="seven">7</button>
    <button id="eight">8</button>
    <button id="nine">9</button>
    <button id="divide">/</button>
  </div>
```



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```
<div>
    <button id="four">4</button>
    <button id="five">5</button>
    <button id="six">6</button>
    <button id="multiply">*</button>
</div>

<div>
    <button id="one">1</button>
    <button id="two">2</button>
    <button id="three">3</button>
    <button id="subtract">-</button>
</div>

<div>
    <button id="clear">C</button>
    <button id="zero">0</button>
    <button id="equals">=</button>
    <button id="add">+</button>
</div>

<script src="script.js"></script>

</body>

</html>
body {
    text-align: center;
    font-family: Arial, sans-serif;
}

#display {
    width: 250px;
    height: 40px;
    font-size: 20px;
    margin-bottom: 10px;
```



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```
}

button {
    width: 55px;
    height: 45px;
    margin: 3px;
    font-size: 18px;
}

let display = document.getElementById("display");

let firstNumber = "";
let operator = "";

function addNumber(number) {
    display.value += number;
}

function setOperator(op) {
    firstNumber = display.value;
    operator = op;
    display.value = "";
}

function calculate() {
    let secondNumber = display.value;
    let result;

    if (operator == "+") {
        result = Number(firstNumber) +
Number(secondNumber);
    }
    else if (operator == "-") {
        result = Number(firstNumber) -
Number(secondNumber);
    }
    else if (operator == "*") {
```



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```
        result = Number(firstNumber) *
Number(secondNumber);
    }
    else if (operator == "/") {
        result = Number(firstNumber) /
Number(secondNumber);
    }

    display.value = result;
}

function clearDisplay() {
    display.value = "";
    firstNumber = "";
    operator = "";
}

document.getElementById("seven").addEventListener("click"
, function() {
    addNumber(7);
});

document.getElementById("eight").addEventListener("click"
, function() {
    addNumber(8);
});

document.getElementById("nine").addEventListener("click",
function() {
    addNumber(9);
});

document.getElementById("four").addEventListener("click",
function() {
    addNumber(4);
});
```



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```
document.getElementById("five").addEventListener("click",
function() {
    addNumber(5);
});

document.getElementById("six").addEventListener("click",
function() {
    addNumber(6);
});

document.getElementById("one").addEventListener("click",
function() {
    addNumber(1);
});

document.getElementById("two").addEventListener("click",
function() {
    addNumber(2);
});

document.getElementById("three").addEventListener("click",
, function() {
    addNumber(3);
});

document.getElementById("zero").addEventListener("click",
function() {
    addNumber(0);
});

document.getElementById("add").addEventListener("click",
function() {
    setOperator("+");
});

document.getElementById("subtract").addEventListener("cli
ck", function() {
```



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```
setOperator("-");
});

document.getElementById("multiply").addEventListener("click", function() {
    setOperator("*");
});

document.getElementById("divide").addEventListener("click", function() {
    setOperator("/");
});

document.getElementById("equals").addEventListener("click", function() {
    calculate();
});

document.getElementById("clear").addEventListener("click", function() {
    clearDisplay();
});

document.addEventListener("keydown", function(event) {

    if (event.key >= "0" && event.key <= "9") {
        addNumber(event.key);
    }

    if (event.key == "Enter") {
        calculate();
    }

    if (event.key == "Escape") {
        clearDisplay();
    }

});
```



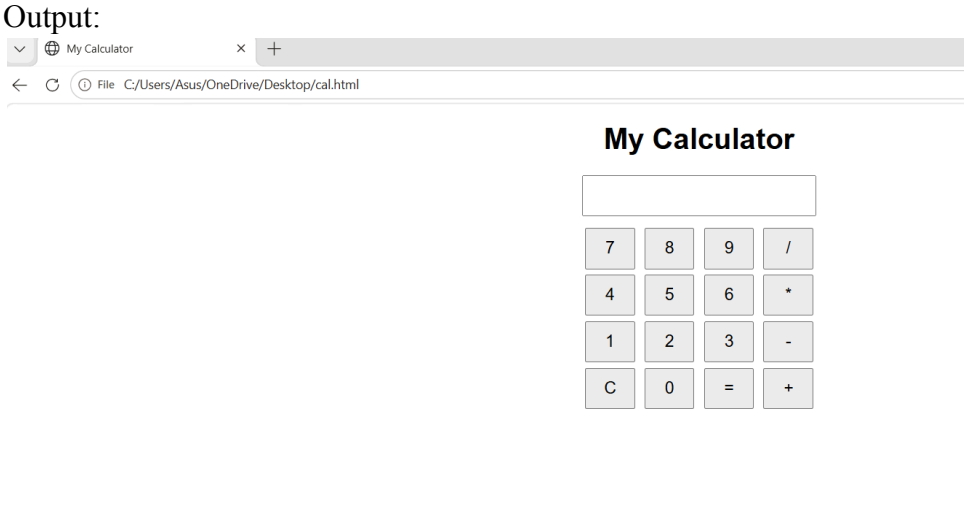
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|                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                       | <p><b>Output:</b></p>                                                                                                                                                                                                                                                                                                                                                                                                  |
| Output Analysis                                                       | The interactive calculator successfully demonstrates JavaScript event handling and DOM manipulation. Button clicks are detected using <code>addEventListener()</code> , and keyboard input is handled using the <code>keydown</code> event. The calculator performs addition, subtraction, multiplication and division using conditional statements. The result is dynamically displayed using the <code>.value</code> property, while the Clear function resets the display, first number and operator. |
| Link of student Github profile where lab assignment has been uploaded | <a href="https://github.com/juhiparwa-1507/Web-Technology-Lab.git">https://github.com/juhiparwa-1507/Web-Technology-Lab.git</a>                                                                                                                                                                                                                                                                                                                                                                          |
| Conclusion                                                            | Thus, an interactive calculator was successfully developed using HTML, CSS and JavaScript. The practical demonstrates event handling, DOM manipulation, functions, conditional statements, arithmetic calculations, button events and keyboard events.                                                                                                                                                                                                                                                   |
| Plag Report (Similarity index < 12%)                                  | Plagiarism Report: To be attached.                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |



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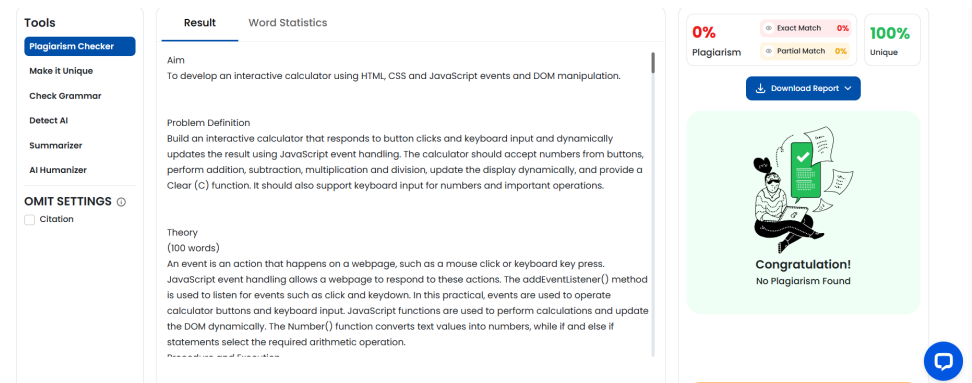
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|      |                                                                                    |
|------|------------------------------------------------------------------------------------|
|      |  |
| Date | 23-09-26                                                                           |